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ReactiveCosmoMapper: A Unified Computational Framework for Entropic Gravity (v1.0.0)

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Abstract

ReactiveCosmoMapper is a high-fidelity *code-first physics* engine designed to validate the Entropic Gravity hypothesis (Verlinde, 2016) as a complete alternative to the ΛCDM model. Unlike standard cosmological simulations that rely on fitted dark matter parameters, this framework computes gravity as an emergent geometric response of spacetime entropy, testing the model consistently across six orders of magnitude in scale.

Key Results and Validation Pillars

- **Galactic scale (≈10 kpc):** Recovered flat rotation curves for SPARC galaxies (e.g., NGC 0024) using purely baryonic mass distributions.
- **Local scale (≈100 kpc):** Demonstrated the spontaneous formation of the *Plane of Satellites* through the External Field Effect (EFE), addressing a long-standing tension in the standard cosmological model.
- **Cluster scale (≈1 Mpc):** Produced “phantom mass” distributions via weak-lensing tomography, reproducing observed cluster lensing signals without invoking non-baryonic matter.
- **Cosmological scale (≈100 Mpc):** Validated the void size function, yielding deeper and cleaner voids consistent with the so-called Peebles tension.
- **Temporal scale (high-z):** Simulated accelerated collapse of primordial gas clouds by introducing a redshift-dependent critical acceleration, offering a natural explanation for the “impossible early galaxies” observed by JWST.

Methodology

The framework implements a custom N-body integrator coupled to a **Reactive Kernel**, responsible for mediating the transition between Newtonian and entropic gravity regimes while incorporating environmental symmetry breaking through the External Field Effect.

Files

ReactiveCosmoMapper_v1.0.0.zip

satellite_plane_collapse.png	100.1 kB
src	
data_ingestor.py	4.8 kB
reactive_cosmo_mapper.py	4.6 kB
reactive_gravity.py	1.9 kB
run_ingestion.py	1.4 kB
run_statistics.py	2.3 kB
solidify_universe.py	4.2 kB
sparc_ingestor.py	3.7 kB
statistics.py	5.4 kB
test_reactive_gravity.py	2.9 kB
visualizer.py	4.1 kB
void_size_distribution.png	46.2 kB
data	
sdss_sample.csv	3.7 MB

Files (12.6 MB)

Additional details

Software

Repository URL

https://github.com/dougdotcon/ReactiveCosmoMapper

Programming language

Python

Development Status

Active

Citations

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Literature (0) | Dataset (0) | Software (0)

Unknown (0) | Citations To This Version

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External resources

Archived in

Software Heritage
swh:1:dir:76440f9977793a67d81277de0bbb8b258a492371

Indexed in

OpenAIRE

Keywords and subjects

Entropic Gravity | Emergent Gravity | Dark Matter | Alternative | Cosmology | N-Body Simulation | Galactic Dynamics | Weak Lensing | Cosmic Voids | JWST | Made

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